

ICAD RT-4 · Set 5 — Answer Sheet

Exam 12-10-2026 · 75 Q · 300 marks · 3 hours · +4 correct, -1 wrong, 0 blank

Name

Roll No

Date

Physics

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Numerical — write the value

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Mathematics

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Numerical — write the value

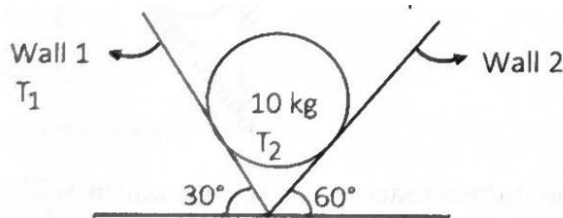
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Bubbles — single correct · Box — write the value · each subject out of 100

Physics

PART A • Q1–20 • SINGLE CORRECT

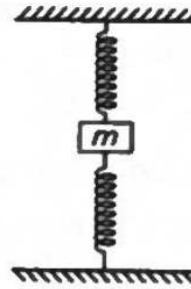
1. P1L2V5Q5 A(3, 2), B(-2, 1) are vertices of triangle ABC with centroid G(5/3, -1/3); find third vertex C. (a) (4, 4) (b) (4, -6) (c) (4, -2) (d) (4, -4)
2. P2L3V5Q5 $F = \sqrt{r(2-r^2)}$. Find dF/dr . (a) $1/\sqrt{r} - 5r^{3/2}/2$ (b) $1/(2\sqrt{r}) - 5r^{3/2}/2$ (c) $-5r^{3/2}/(2\sqrt{r})$ (d) $5/2$
3. P3L2V3Q4 The vectors A and B are such that $A+B=C$ and $A^2+B^2=C^2$. If θ is the angle between positive directions of A and B then mark the correct alternative: (a) $\theta=0^\circ$ (b) $\theta=\pi/2$ (c) $\theta=2\pi/3$ (d) $\theta=\pi$
4. P3L2V6Q8 (Multiple Correct) For two fixed magnitudes A and B, the resultant $R(\theta)=\sqrt{A^2+B^2+2AB \cos\theta}$ varies with θ . (a) R is maximum at $\theta=0$ and minimum at $\theta=\pi$ (b) $dR/d\theta=-AB \sin\theta/R$; slope is zero only at $\theta=0, \pi$ (c) Near $\theta=\pi$, $R \approx |A-B| + (AB/2|A-B|)(\pi-\theta)^2$ (positive curvature) (d) If $A=B$, then $R(\theta)=2A \cos(\theta/2)$ for all θ
5. P3L3L4V4Q4 $A = 3\hat{i} + 4\hat{j}$, $B = 7\hat{i} + 24\hat{j}$: vector with same magnitude as B, parallel to A. (a) $15\hat{i} + 20\hat{j}$ (b) $3\hat{i} + 4\hat{j}$ (c) $7\hat{i} + 24\hat{j}$ (d) None
6. P4L1V2Q3 A body covered a distance of 5 m along a semicircular path. The ratio of distance to displacement is: (a) 11:7 (b) 12:5 (c) 8:3 (d) 7:5
7. P4L2V6Q10 (Multiple correct) For constant a, compare S_n (displacement in nth second) and average velocity over $[0, t]$: (a) $S_n = u + (1/2)a(2n-1)$ (b) $\bar{v} = s/t = (u+v)/2$ (c) if $a < 0$, S_n must be negative for large n (d) if $u=0$, S_n is an arithmetic progression in n
8. P4L4V5Q3 A balloon is moving vertically with a velocity of 4 ms^{-1} . When it is at a height of h, a body is gently released from it. If it reaches the ground in 4 seconds, the height of the balloon when the body is released is: (a) 62.4 (b) 42.4 (c) 78.4 (d) 82.2 m
9. P4L7V3Q3 A projectile has $u = 25 \text{ m/s}$ and covers a range of 54 m. Find the angle of projection: (a) 60° (b) 30° (c) 45° (d) 53.1°
10. P5L2V4Q9(i) A 10 kg sphere rests between Wall 1 (30°) and Wall 2 (60°): (i) normal force between Wall 1 and sphere: (a) $50\sqrt{3}$ (b) $100\sqrt{3}$ (c) $50/\sqrt{3}$ (d) $100/\sqrt{3}$



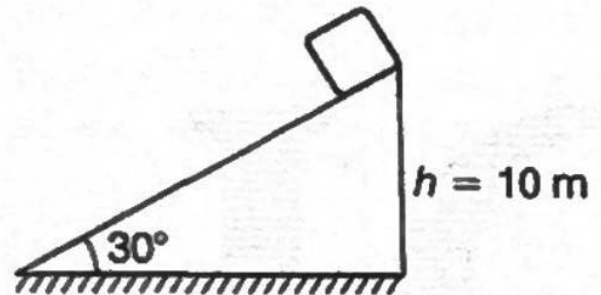
11. P6L2V3Q5 Three blocks 4 kg, 8 kg, 24 kg by light strings on smooth floor, $a=2 \text{ ms}^{-2}$. Applied force F. (a) 36 N (b) 72 N (c) 18 N (d) 108 N



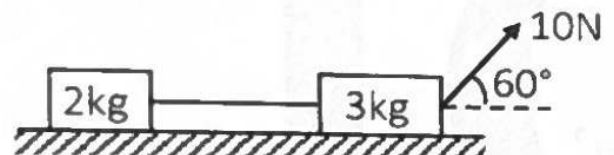
12. P6L5V4Q3 Block between two springs in equilibrium; upper cut $\rightarrow 6 \text{ m/s}^2$ down. If lower cut, magnitude of acceleration ($g = 10$)? 16 / 4 / Cannot determine / None



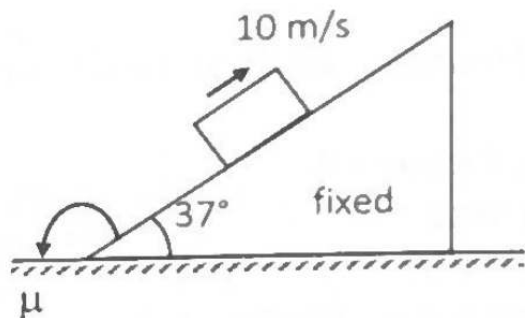
13. P7L2V3Q5 Consider a car moving along a straight horizontal road with a speed of 72 km/h. If the coefficient of static friction between the tyre and the road is 0.5, the shortest distance in which the car can be stopped is (take $g = 10 \text{ m s}^{-2}$): (a) 30 m (b) 40 m (c) 72 m (d) 20 m
14. P8L1V5Q6 A block of mass 5 kg slides down an inclined plane of inclination 30° from a height of 10 m from the ground. The coefficient of friction between the block and the inclined plane is 0.5. Find the work done by friction on the block. (a) $250\sqrt{3} \text{ J}$ (b) $-250\sqrt{3} \text{ J}$ (c) 500 J (d) -500 J



15. P8L2V3Q2 A body of mass 5 kg at rest is under the action of a force which gives it a velocity given by $v = 3t \text{ m/s}$, here 't' is time in seconds. The work done by the force in two seconds will be (a) 90 J (b) 45 J (c) 180 J (d) 30 J
16. P8L5V3Q4 The potential energy for a body of mass m that is acted on by a very massive body is given by $U = -mgx + (1/2)kx^2$. The corresponding force is: (a) $-mgx^2/2 + kx^3/6$ (b) $mg - kx$ (c) $-mg + kx/2$ (d) $-mg + kx$
17. P8L8V5Q8 A machine gun fires 420 bullets per minute. The velocity of each of the bullet is 300 ms^{-1} and the mass of each bullet is 1 gm. The power of machine gun is (a) 315 W (b) 315000 W (c) 630 W (d) 3150 W
18. P5L2V5Q8 A 2 kg block is connected by a string to a 3 kg block, which has a 10 N force applied at 60° ; find the tension in the connecting string: (a) 2 N (b) 3 N (c) 5 N (d) 10 N

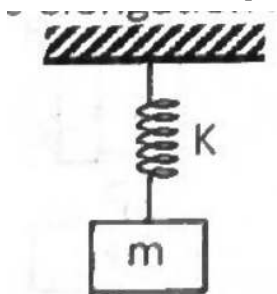


19. P7L3V3Q4 Find the distance travelled by a block on an incline before it stops. The block moves up a fixed 37° incline with initial velocity 10 m/s and the coefficient of friction between block and incline is $\mu = 0.5$. (a) 2 m (b) 5 m (c) 3 m (d) 4 m ($g = 10 \text{ m/s}^2$)

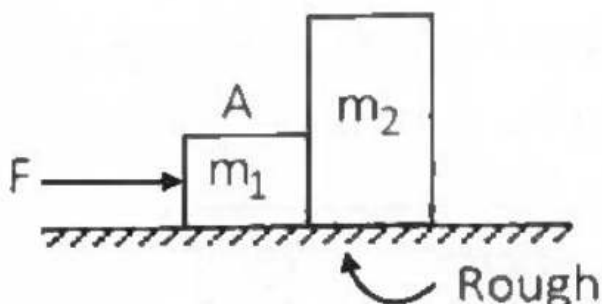


PART B · Q21–25 · NUMERICAL VALUE

21. P4L5V6Q4 A bird flies for 4s with velocity $v = (t-2) \text{ m/s}$ in a straight line. Calculate the distance covered by the bird (in meter)
22. P6L5VSuggQ1 A block of mass m hangs from a string of spring constant k ; block at equilibrium. Find the elongation.

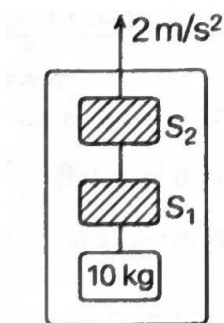


23. P7L1V4Q7 Find the number of forces in the F.B.D. of block 'A': a short block m_1 (labelled A) on rough ground is pushed by a horizontal force F , and is in contact with a taller block m_2 that also stands on the same rough ground.



20. P8L6V5Q1 A 15 g ball is shot from a spring gun whose spring has a force constant of 600 N/m . The spring is compressed by 5 cm. The greatest possible horizontal range for this compression is ($g = 10 \text{ m/s}^2$) (a) 6.0 m (b) 12.0 m (c) 10.0 m (d) 8.0 m

24. P4L5V4Q1 A particle moves along a straight line such that its displacement $s = t^3 - 6t^2 + 3t + 4$ meters. Find the velocity (in m/s) when the acceleration is zero
25. P6L6V3Q5 All spring balances light, steady state. Find readings (S_1 and S_2): $S_2/S_1/10 \text{ kg}$ accelerating upward at 2 m/s^2



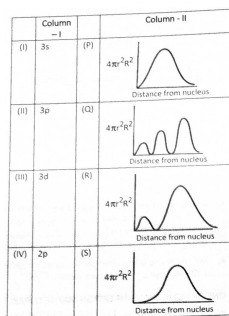
Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C1L5V3Q2 H_2 diffuses in 4 s — time for O_2 ($M=32$). (a) 16 s (b) 4 s (c) 8 s (d) 11.3 s
2. C1L6V4Q3 27°C , 1 atm; pressure doubled & $T \rightarrow 327^\circ\text{C}$ — new avg KE E_2 vs E_1 per molecule? (a) $E_2 = 2E_1$ (b) $E_2 = 3E_1$ (c) $E_2 = E_1$ (d) $E_2 = 1/2 E_1$
3. C1L8V5Q4 Equal O_2 and H_2 molecules, same T & P — correct statement on real-gas behaviour. (a) Both have identical compressibility factors (Z) at all pressures. (b) H_2 shows smaller deviation from ideal behavior due to weaker van der Waals forces. (c) O_2 shows smaller deviation because of larger molecular size. (d) Z for both gases is always less than 1 at all pressures.
4. C2L6V3Q3 $T_2 = 3.0 \times 10^{-16} \text{ s} \rightarrow$ time period in 4th orbit (T_4). (a) $2.4 \times 10^{-15} \text{ s}$ (b) $3.75 \times 10^{-16} \text{ s}$ (c) $1.2 \times 10^{-15} \text{ s}$ (d) $6.0 \times 10^{-15} \text{ s}$

5. C2L6V4Q9 Hydrogen-like ion $Z=4 \rightarrow$ speed of its electron in 2nd orbit. (a) $8.72 \times 10^6 \text{ m/s}$ (b) $4.36 \times 10^6 \text{ m/s}$ (c) $2.18 \times 10^6 \text{ m/s}$ (d) $1.09 \times 10^6 \text{ m/s}$
6. C2L9V4Q2 Any nf-orbital can accommodate up to (a) 14 electron (b) Six electrons (c) Two electrons with parallel spin (d) Two electrons with opposite spin

7. C2L11V5Q2 Column-match: $3s / 3p / 3d / 2p \rightarrow 4\pi r^2 R^2$ plots. (a) (I)-(S), (II)-(P), (III)-(Q), (IV)-(R) (b) (I)-(P), (II)-(Q), (III)-(S), (IV)-(R) (c) (I)-(R), (II)-(Q), (III)-(P), (IV)-(S) (d) (I)-(Q), (II)-(R), (III)-(S), (IV)-(P)



8. C3L4V3Q2 E.F. = CH, molecular mass = 78. Find M.F. (a) CH (b) C₄H₄ (c) C₆H₆ (d) C₃H₃
9. C3L5V3Q5 How many litres of NH₃ are formed from 28 g of N₂ at STP? (N₂ + 3H₂ → 2NH₃). (a) 22.4 L (b) 44.8 L (c) 33.6 L (d) 11.2 L
10. C3L6V4Q4 10 mL of 1 M BaCl₂ + 5 mL of 0.5 M K₂SO₄ → BaSO₄. Amount of BaSO₄ precipitated. (a) 0.005 mol (b) 0.00025 mol (c) 0.025 mol (d) 0.0025 mol
11. C4L5V5Q9 Among the following, the incorrect electron gain enthalpy order is (compare magnitude only): (a) Li > H > Na > K (b) Cl > F > Br > I (c) S > Se > Te > Po > O (d) Cl > F > S > O
12. C4L6V6Q5 Aqueous solution of two compounds M₁—O—H and M₂—O—H are prepared in two different beakers. If the electronegativity of M₁ = 3.4, M₂ = 1.2, O = 3.5 and H = 2.1, then the nature of two solutions will be respectively: (a) acidic, basic (b) acidic, acidic (c) basic, acidic (d) basic, basic

PART B · Q21–25 · NUMERICAL VALUE

21. C3L6V5Q1 130 g HCl added to 130 g zinc. Volume of H₂ gas evolved (STP) [Zn = 65]
22. C4L5V4Q2 Among the following elements, what is the atomic number of the one which possesses zero electron affinity — N, O, Ne, Li?
23. C3L5V4Q4 11.2 gram iron can be obtained from complete reduction of how many grams of ferric oxide (Fe₂O₃)? (Fe = 56, O = 16)

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M1L2V5Q8 Dial hand 6 cm; swept-area statements (multi). (a) The minute hand sweeps area $12\pi \text{ cm}^2$ in 20 minutes. (b) The minute hand sweeps area $12\pi \text{ cm}^2$ in 10 minutes. (c) The hour hand sweeps area $12\pi \text{ cm}^2$ in 240 minutes. (d) In 60 minutes, the minute hand sweeps area $36\pi \text{ cm}^2$.
2. M2L1V6Q2 $A = P(\{1,2\})$; correct statements (multi-correct). (a) $\emptyset \in A$ (b) $\{1\} \subset A$ (c) $\{1,2\} \in A$ (d) $1 \in A$
3. M3L2V5Q4 $(2p+1)\sin\alpha = 2p(p+1)\cos\alpha \rightarrow \sin\alpha$. (a) $(2p+1)/(2p^2 + 2p + 1)$ (b) $(2p+3)/(2p^2 + 2p + 1)$ (c) $2p(p+1)/(2p^2 + 2p + 1)$ (d) 1
4. M3L3V5Q5 #solutions of $\tan(270^\circ - x) = \cot(x + 180^\circ)$ in $[0^\circ, 360^\circ)$. (a) 0 (b) 2 (c) All x in $[0^\circ, 360^\circ)$ where both sides are defined (d) Exactly 4
5. M4L2V5Q5 Domain of $f(x) = (x-2)/(x^2-1)$. (a) $x \neq \pm 1$ (b) $x \neq 1, x > 2$ (c) $x \in \mathbb{R} \setminus \{-1, 1\}$ (d) $x > 1, x \neq 2$
6. M4L4V2Q1 Period of $\cos(2x)$. (a) π (b) 2π (c) $\pi/2$ (d) 4π

13. C5L1V4Q7 An ionic bond is more likely to form when the electronegativity difference is large because: (a) Both atoms tend to share electrons (b) Both atoms have similar electron affinities (c) One atom loses electrons easily and the other gains electrons easily (d) Both atoms resist electron transfer
14. C5L1V5Q6 Elements whose electronegativities are 1.2 and 3.0, form: (a) ionic bond (b) covalent bond (c) coordinate bond (d) metallic bond
15. C5L5V4Q2 What is common between the following molecules: SO₃, CO₃(²⁻), NO₃(⁻)? (a) All have linear shape (b) All have trigonal planar shape (c) All have tetrahedral shape (d) All have trigonal pyramidal shape
16. C5L7V3Q4 Which statement correctly differentiates atomic and molecular orbitals? (a) Atomic orbitals are polycentric (b) Molecular orbitals are influenced by only one nucleus (c) Atomic orbitals have complex shapes (d) Molecular orbitals are influenced by two or more nuclei
17. C5L9V5Q6 Among the following, the molecule with the highest dipole moment is: (a) CH₃Cl (b) CH₂Cl₂ (c) CHCl₃ (d) CCl₄
18. C1L9V4Q9 3 mol SO₂ occupy 10 L at 15 atm — find T, $a=6.71$, $b=0.0564$. (a) 523 K (b) 623 K (c) 672 K (d) 741 K
19. C2L9V3Q2 How many electrons can have $n=3$ and $\ell=2$? (a) 6 (b) 10 (c) 14 (d) 8
20. C5L4V3Q1 In NH₃, there are 3 bond pairs and 1 lone pair. What is Q (total number of orbitals) and hybridisation? (a) Q=4, sp³ (b) Q=3, sp² (c) Q=4, sp² (d) Q=3, sp³

24. C4L6V5Q5 Calculate the electronegativity of fluorine atom from the following data : E H—H = 104.2 Kcal/mol, E F—F = 36.6 Kcal/mol, E H—F = 134.6 Kcal/mol, $\chi \text{ H} = 2.1$
25. C1L2V6Q3 1.00 L at unknown P; +0.96 atm → 0.25 L, then isothermal to 0.20 L; final pressure. (a) 1.50 atm (b) 1.60 atm (c) 2.00 atm (d) 0.80 atm
7. M5L2V5Q9 If $5 \sin\alpha = 3 \sin(\alpha + 2\beta) \neq 0$, then $\tan(\alpha + \beta)$ is equal to (a) $2 \tan\beta$ (b) $3 \tan\beta$ (c) $4 \tan\beta$ (d) $6 \tan\beta$
8. M5AL3V6Q6 If $3 \tan A - 4 = 0$ where A lies in the 3rd quadrant, then $\cos(A + \pi/3) = ?$ (a) $(4\sqrt{3}+3)/10$ (b) $(4\sqrt{3}-3)/10$ (c) $(2\sqrt{3}+4)/10$ (d) $(\sqrt{3}+5)/10$
9. M5CL4V4Q9 $A + B + C = 180^\circ \Rightarrow \Sigma (\cot A + \cot B)/(\tan A + \tan B) =$ (a) -1 (b) 0 (c) 1 (d) 2
10. M6L2V5Q9 If $(2x-3)/(x-1) + 1 = (6x-x^2-6)/(x-1)$, then 'x' is: (a) 1 (b) -2 (c) 2 (d) 4
11. M6L3V3Q1 If $|2x-5| = 7$, then x is: (a) -1, 6 (b) 6, 5 (c) 12, 2 (d) None of these
12. M7L3V3Q4 If a, b, c are three consecutive positive integers, then $\log(1+ca) =$ (a) $\log b$ (b) $\log(b/2)$ (c) $\log(2b)$ (d) $2 \log b$
13. M7L3V5Q8 If $(a^{(\log_b x)})^2 - 5x^{(\log_b a)} + 6 = 0$, where $a>0$, $b>0$ and $ab \neq 1$, then the value of x can be equal to: (a) $2^{(\log_b a)}$ (b) $3^{(\log_a b)}$ (c) $b^{(\log_a 2)}$ (d) $a^{(\log_b 3)}$

- 14.** M8AL2V4Q5 If the sum of the 10 terms of an A.P is 4 times to the sum of its 5 terms, then the ratio of first term and common difference is (a) 1 : 2 (b) 2 : 1 (c) 2 : 3 (d) 3 : 2
- 15.** M8BL1V5Q2 If n th terms of 5, $-5/2$, $5/4$, $-5/8$, is $5/1024$, then value of n is (a) 11 (b) 10 (c) 9 (d) 12
- 16.** M8CL1V5Q10 If 'a' and 'b' are the first and the last terms of an A.P. as well as of an H.P., both having n terms, then the product of the r th term of the first series and the $(n - r + 1)$ th term of the second series is (a) independent of both n and r and equals ab (b) independent of both n and r and equals a^2b (c) independent of both n and r and equals ab^2 (d) dependent on n and r

PART B · Q21–25 · NUMERICAL VALUE

- 21.** M5AL2V4Q6 Find the value of $(\tan 70^\circ - \tan 20^\circ)/\tan 50^\circ$.
- 22.** M6L4V5Q3 If solution set of $3x^2 - 7x + 4 \leq 0$ is $[a, \lambda/\mu]$ where HCF of λ, μ is 1, then $\lambda \cdot \mu + a$ is _____
- 23.** M2L3V3Q1 $A = \{1..7\}$, number of proper subsets = K $\rightarrow$ product of digits of K

- 17.** M3L2V5Q8 $(\cos^3 A + \sin^3 A)/(\cos A + \sin A) + (\cos^3 A - \sin^3 A)/(\cos A - \sin A) = k$. (a) 0 (b) 1 (c) 2 (d) -1
- 18.** M7L4V5Q5 If $\log 3 = 0.4771$, then the position of the first significant digit in the value of $3^{(-20)}$ is: (a) 10th (b) 11th (c) 12th (d) 13th
- 19.** M8AL1V3Q4 If T_r is the r th term of an A.P. and if $T_3 = 1/6$ and $T_6 = 1/3$, then find the ratio of the first term to the common difference. (write the value)
- 20.** M8BL1V3Q4 Find 4 th term from last in sequence $2/27, 2/9, 2/3, \dots\dots\dots 162$. (write the value)

- 24.** M5CL1V3Q2 If $\sin A = 3/5$, $0^\circ < A < 90^\circ$, then the value of $\tan 2A$ is:
- 25.** M4L2V3Q1 Domain of $f(x) = \sqrt{x-1}$ is $[a, \infty) \rightarrow a+5$